



LWSNet: A lightweight network for automated welding point cloud segmentation

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ABSTRACT

Welding technology is a key manufacturing link in modern industrial steel structures, directly impacting welding quality and efficiency. This paper proposes LWSNet, a novel point cloud part segmentation network aimed at accurately segmenting weld areas across various joints to enhance automation and ensure quality. LWSNet builds upon the PointNet++ model, incorporating depthwise separable convolution and normalization-based attention module in its feature extraction block (FEB) for improved lightweight and feature extraction. A linear bottleneck structure and adaptive maximum pooling in its feature generation block (FGB) significantly reduce information loss during feature transmission. Additionally, a laser vision sensor model is designed for point cloud data acquisition, and the WeldJoint-PCD dataset is constructed using multiple data augmentation methods. Experimental results on WeldJoint-PCD show LWSNet's superior segmentation effectiveness and lightweight design. On ShapeNetPart, cls.mIoU and ins.mIoU reach 82.9% and 85.5%, respectively, outperforming the original model by 1% and 0.4%, demonstrating LWSNet's strong generalization capabilities.

1. Introduction

With the development of modern industry, the quality and performance requirements of steel structures are getting higher and higher, and welding, as a key link in manufacturing steel structures, has a direct impact on the overall performance of steel structures in terms of their quality and efficiency. With the development of automation and intelligent technology, welding technology is also progressing, and robot welding is gradually replacing manual welding [1,2]. At present, the basic form of welding joints commonly used in the welding process can be divided into four types of T joints, lap joints, butt joints and corner joints [3,4], and their welding requirements are shown schematically in Fig. 1. The selection of the form of welded joints has an important impact on the mechanical properties of the welded structure and manufacturing process. Therefore, it is of great significance to accurately segment the point cloud of the weld area in different welded joints to improve the level of welding automation and ensure welding quality and efficiency.

However, most robots still remain in the fixed mode of manual teach-back in welding operations, which is cumbersome and error-prone [5]. In industrial field weld seam measurement, a laser of suitable frequency is usually chosen as the light source of the vision sensor [6–10], and this approach can accurately capture the workpiece surface information. To accurately localize the characteristic points of a weld

seam, researchers have explored several methods, most of which are based on image processing techniques. Although these methods can achieve high accuracy in some cases, they are often limited to specific weld types and lose global point cloud information. In addition, methods based on image processing can only work effectively in weak noise environments and have limited effectiveness in harsher industrial environments. Compared to image processing techniques, point cloud processing techniques provide complete three-dimensional information that captures the geometric features of the weld seam in a more comprehensive way, which is crucial for the identification and localization of different welded joints. At the same time, point cloud processing technology is less affected by external conditions and can maintain high stability and accuracy in industrial environments with strong noise and dynamic changes, and point cloud processing technology can provide higher flexibility and scalability, which is easy to integrate with modern robotic systems to realize more efficient automated welding. With the development of point cloud segmentation techniques, more and more researchers have begun to apply them to weld area point cloud feature extraction [11–18].

Although traditional point cloud segmentation networks can realize the segmentation of feature point clouds to a certain extent, it is often difficult to meet the real-time requirements in actual production due to

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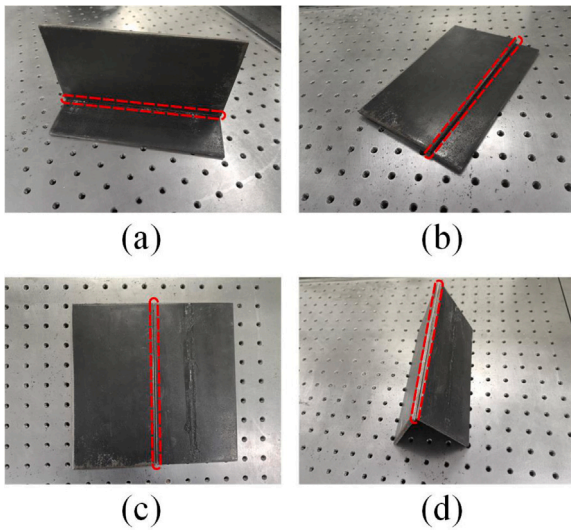


Fig. 1. Schematic diagram of welding requirements for commonly used welded joints. (a) T joints. (b) Lap joints. (c) Butt joints. (d) Corner joints.

their complex network structure and large number of parameters. To quickly and accurately segment the weld area point clouds in different welded joints for subsequent path planning and guiding the robot to carry out the welding work, this paper proposes a novel point cloud part segmentation network model LWSNet (Lightweight weld segmentation network) based on PointNet++ [19]. The network mainly consists of the feature extraction block (FEB), the feature generation block (FGB) and the feature propagation block (FPB), where the processing of the point cloud by the FEB and FGB is called point cloud downsampling, and the processing of the point cloud by the FPB is called point cloud upsampling. Specifically, given an original weld joint workpiece point cloud, FEB is used to sample and group the points in the point cloud and extract the feature information of each point in the point cloud and the position information relative to the sampling point. FGB is used to extract and fuse the information extracted by FEB to obtain the global feature information of the fused local features. Specifically, given an original welded joint workpiece point cloud, FEB is used to sample and group the points in the point cloud and extract the feature information of each point in the point cloud as well as the positional information relative to the sampled points, and FGB is used to extract the local point cloud features from the information extracted by FEB and fuse them to obtain the global feature information that fuses the local features, and FPB fuses the high-dimensional feature information obtained from the point cloud downsampling with the original feature information, improving the expression ability of the point cloud features. Finally, the fused feature information of the point cloud obtained from upsampling is passed through the network detection head LWS Head to classify each point and get the point cloud segmentation results.

In FEB, we introduce the depthwise separable convolution and normalization-based attention module [20,21], which can effectively reduce the number of parameters and computation amount of the network model and help the network model to obtain the salient features in the point cloud. At the same time, FGB can effectively solve the gradient vanishing problem in the network model, reduce the information loss of point cloud feature information when it is transmitted in different dimensional spaces and improve the feature extraction ability of the network model. To effectively train our network model and enable it to be applied to the point cloud segmentation task in the weld area, we designed a laser vision sensor model to realize the point cloud data acquisition of the weld joint workpiece, and constructed a welded joint workpiece point cloud dataset WeldJoint-PCD through the traditional data augmentation method and the point cloud augmentation

with weighted local transformations (PointWOLF) [22]. The experimental results show that our network model has a better Lightweight effect compared to the more popular point cloud part segmentation network models, and has the best point cloud segmentation effect on the WeldJoint-PCD dataset, while showing good generalization ability on the ShapeNetPart dataset [23]. In summary, the main contributions of this work can be summarized as follows:

1. We propose a novel point cloud part segmentation network model that can quickly and accurately segment the weld area point clouds in different welded joints.
2. We design the FEB module, which significantly reduces the number of parameters and computation of the network model and improves the feature extraction capability of the network model by introducing depthwise separable convolution and NAM attention mechanism.
3. For the possible gradient vanishing problem in the feature extraction part of the point cloud, we design the FGB module, which reduces the information loss in the transmission of the point cloud feature information and improves the feature generation capability of the network model.
4. We designed a laser vision sensor model to realize the acquisition of point cloud data, and constructed WeldJoint-PCD, a point cloud dataset of welded joint workpieces, through multiple data augmentation methods, which serves as the basis for network model training, validation, and inspection.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the laser vision sensor model we designed and the construction of the WeldJoint-PCD dataset. Section 4 details our proposed LWSNet network model. Section 5 reports our experimental results and analyzes them. Finally, conclusion is given in Section 6.

2. Related work

2.1. Image processing based methods

In the field of welding, to localize the feature points of weld seams, researchers have explored a variety of methods, and the traditional weld feature point localization methods mainly rely on image processing techniques. Zou and Zeng [7] proposed a lightweight segmentation network based on SOLOV2, which is capable of accurately segmenting the laser stripes under strong noise. He et al. [24] proposed a scheme to extract weld profile feature points for automated multi-pass route planning and to guide the initial weld position of each layer during metal active gas arc welding. Cheng et al. [25] proposed an effective method for extracting feature points from the weld seam images in noisy construction environments, and this method redefined the feature point extraction as a skeleton critical point detection task. Chen et al. [26] proposed a symmetric encoder-decoder semantic segmentation model LSRNet, which realizes the real-time detection of laser streaks under the strong interference environment of electric arcs and analyzes the difference between arcs and laser streaks from the semantic point of view. He et al. [27] investigated a Bayesian network model (BNM) to implement the autonomous decision-making process of welding positions in gas metal arc multipass welding with T-joints for automated manufacturing. The experimental results showed that their proposed BNM can give the effective decision-making results of welding positions with T-joints of different thicknesses and show great potential for higher manufacturing efficiency and automatic levels. Banafian et al. [28] developed an integrated robot trajectory extraction method for complex weld seams by processing noisy images and laser data. Johan et al. [29] developed a qualitative description technique based on buttressed V-grooves by proposing a method to improve the quality of the centerline by constructing a more stable centerline, in response to the problem that the noise generated by the laser beam on

the surface of the workpiece can cause the centerline of the laser to become inhomogeneous leading to difficulties in extracting the feature points of the weld. He et al. [30] proposed an autonomous decision and optimization method of welding positions to implement angular distortion and weld collapse control during the arc welding process of thick plate T-joints with a K-groove, via the analytic hierarchy process (AHP), which is a valuable solution for improving manufacturing efficiency and automation in the field of arc welding of thick plates. Wang et al. [31] proposed a practical and robust weld tracking algorithm, which takes the problem of weld feature point detection as the detection and tracking of laser streaks, and the real-time accuracy of the algorithm was proved by comparison experiments with other methods. Although these methods can achieve a certain level of accuracy in some specific cases, they are usually limited by the diversity of image quality, lighting conditions, and weld types, leading to their robustness and generalization in complex industrial environments. He et al. [32] proposed a unified framework based on a modified U-Net for extracting Weld seam profiles with various joints regardless of the position, orientation and shape of the Weld seam profiles in the image. It shows the potential for automated and intelligent arc welding.

2.2. Point cloud processing based method

In recent years, point cloud processing based methods have shown significant advantages in weld feature point localization [33]. Compared with the traditional image processing-based methods, the point cloud processing-based methods can provide more accurate and comprehensive 3D spatial information, which results in higher accuracy and robustness in the localization of weld feature points. The introduction of point cloud data allows the system to work in complex environments, including light variations and undesirable surface conditions of the workpiece. He et al. [24] proposed a weld extraction method based on a point cloud processing algorithm for the problems of inefficient wire structured light and poor robustness of passive vision. The algorithm is well adapted to weld seams of different types and groove sizes. Yang et al. [11] proposed a 3D weld seam extraction algorithm based on point cloud segmentation, according to the space structure of weld seams, the 3D seam path model and pose estimation are solved based on the established mathematical model of weld seams. Zhou et al. [13] proposed a point cloud-based arc welding robot system, in which the intensity-based edge point detection algorithm used in the system has a high level of robustness and effectiveness in automated path planning welding applications. Yuankai et al. [33] proposed a 3D vision-based planar weld extraction method using planar V-butt, planar I-butt and planar lap welds as an example, which can well solve the problem of automatic weld path planning for planar V-butt, planar I-butt, and planar lap welds. Ahmed et al. [34] proposed a new unorganized point cloud edge and corner point detection algorithm. A new corner point detector is proposed in this algorithm that clusters the curvature vectors and uses their geometric statistics to classify the points as corners. Kim et al. [35] proposed a new method for the extraction of weld seams with different viewpoint directions based on point cloud alignment, which can deal with randomly occluded weld seams in workpieces with multiple seam parts. Feng et al. [36] propose a novel feature-guided trajectory generation method based on point cloud data to perform an efficient grinding process for welds on a freeform surface.

PointNet [37] is a classical network in the field of point cloud segmentation, but PointNet cannot well extract the local fine features of point clouds and cannot be applied to the recognition tasks that need to identify fine features. To solve this problem, a hierarchical structure is added to PointNet++, which in turn extracts point cloud features at different scales, and experiments have shown that the PointNet++ network can process point sets efficiently and robustly. The design of PointNet++ allows for the flexibility of adding or modifying modules to

suit different application requirements. At the same time, PointNet++ is still computationally efficient even when compared to the more popular current network models for point cloud processing, and is suitable for scenarios that require real-time processing. In the design of deep learning network models, restructuring the structure of an existing network model, adding some specific modules, and optimizing a certain module of the network model may make the network model show better performance. Vanian et al. [38] used Point Transformer on SPVCNN networks to demonstrate that it can make the architecture outperform the Street3D data set outperforming the state-of-the-art at the time. Li et al. [39] proposed X-transformation and combined it with traditional convolution operations to form the X-conv module, which improves the convolution operation for point clouds, allowing the network model to achieve performance comparable to or better than the state-of-the-art at the time on a number of challenging benchmark datasets and tasks. Szegedy et al. [40] proposed the Inception-v4 network, demonstrating that the training of Inception can be significantly accelerated and the performance of the Inception network can be improved by adding residual connectivity. Brauwers and Frasnica [41] provided a comprehensive overview of attention mechanisms in deep learning, demonstrating a variety of ways to enhance the performance of the model by introducing an attention module. Inspired by these approaches, we propose a novel lightweight weld area point cloud segmentation network model based on the PointNet++ network model, which is capable of quickly segmenting weld area point clouds of different welded joints.

3. Laser vision sensors and datasets

3.1. Laser vision sensor measurement model

To obtain the three-dimensional point cloud data of the welded joint workpiece, this paper is based on the binocular line laser scanning imaging method designed laser vision sensor. Binocular line laser scanning system has the advantages of simple construction and easy maintenance, fast detection speed, strong resistance to external light interference ability, etc., when scanning the line laser irradiation on the surface of the measured object, the two cameras are responsible for the acquisition of stripe images. The imaging model of the laser vision sensor is shown in Fig. 2.

After the two cameras are calibrated and co-calibrated, the internal and external parameters of the two cameras, the camera focal lengths f_1 and f_2 , and the rotational translation matrix M_{lr} of the transformation relationship between the two cameras are obtained. 2D coordinates of the line laser beam irradiated on the image are firstly derived, and then 3D coordinates of the welded joint workpiece are calculated according to the principle of binocular imaging. Assuming that a point P on the welded joint workpiece has the corresponding horizontal coordinates x_l and x_r under the left and right camera image coordinate systems $O_{ul} - X_{ul}Y_{ul}$, $O_{ur} - X_{ur}Y_{ur}$, respectively, and that its vertical coordinates correspond one-to-one in the left and right camera images after binocular correction along the pole line L , then the vertical coordinates under the two image coordinates of point P are equal, that is $y_l = y_r$. By the trigonometric geometric relationship can be obtained as follows :

$$x_l = f \frac{x_c}{y_c}, x_r = f \frac{x_c - L}{z_c}, y_l = y_r = f \frac{y_c}{z_c} \quad (1)$$

From this, the 3D coordinates of point P under the camera coordinate system can be calculated, and thus the 3D point cloud data of the welded joint workpiece can be obtained, and the 3D coordinates of point P are calculated as:

$$x_c = \frac{L \cdot x_l}{D}, y_c = \frac{L \cdot y_l}{D}, z_c = \frac{L \cdot f}{D} \quad (2)$$

where the parallax $D = x_l - x_r$.

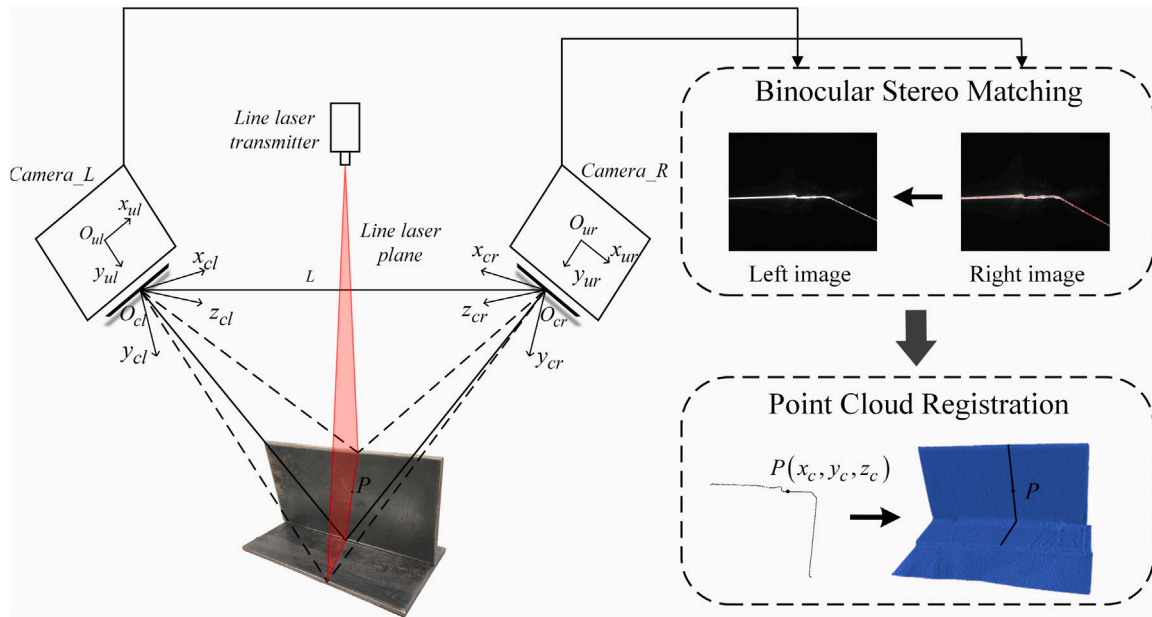


Fig. 2. Imaging model of the laser vision sensor, where P is a point on the welded joint workpiece (T joint as an example).

Table 1

Data augmentation.

Methods	Value
Random translation	shift_range = 0.1
Random rotation	angle_sigma = 0.06, angle_clip = 0.18
Random scaling	scale_low = 0.8, scale_high = 1.25
Random jitter	sigma = 0.13, clip = 0.18, jitter_fraction = 0.0005
PointWOLF	w_num_anchor = 4, w_sample_type = fps, w_siglab = 0.7, w_R_range = 8, w_S_range = 3, w_T_range = 0.5

3.2. Welded joint workpiece dataset

3.2.1. Data augmentation

Data augmentation can effectively solve the problem of experimental data scarcity, and in the deep network model structure can accelerate the convergence speed of the network model training, reduce overfitting and improve the performance of the network model [42]. To fully exert the potential of the dataset, this paper uses the traditional point cloud data augmentation method and the point cloud augmentation with weighted local transformations (PointWOLF) [22] to process the original point cloud collected by the laser vision sensing system to produce the experimental dataset. The specific operations of point cloud data augmentation are shown in Table 1.

The random translations, rotations, scaling, and jittering performed by traditional point cloud data augmentation methods when performing data augmentation on point cloud samples perform the same operation on all points in the point cloud sample, which can simulate a certain degree of variation that occurs in the welded joint artifacts in a real environment. However, the point cloud data samples generated by traditional point cloud data augmentation rely on an imposed global transformation matrix or vector for each original sample and do not take into account the low-local structure of each sample. Therefore, we further use PointWOLF to generate some realistic, locally transformed different point cloud samples. PointWOLF proposes a local weighted transformation centered on multiple anchor points to produce smoothly varying non-rigid deformations. Firstly, several anchor points in the point cloud data are selected by farthest point sampling, and then some

local transformations of random degree are performed around each anchor point, including scaling, changing aspect ratio, translation and rotation. Finally, a kernel regression is performed on the original point cloud data, the anchor point data, and the locally transformed data to generate the enhanced point cloud data.

By using these two point cloud data augmentation methods simultaneously, the stability of the network model can be enhanced in the face of various changes in the point cloud data, helping the model to learn richer and more robust feature representations and helping to improve the generalization ability of the network model on unknown point cloud data.

3.2.2. Construction of the dataset

In this paper, the research object is four common welded joint workpiece point clouds, which have the characteristics of small size, scale invariance, translation invariance and rotation invariance, etc. Combined with the needs of point cloud feature extraction in our weld area, we select the ShapeNet Part dataset released by Stanford University as a reference to produce four kinds of welded joint workpiece point cloud datasets, WeldJoint-PCD, which is composed of a mixture of real work environment acquisition and point cloud data augmentation. WeldJoint-PCD consists of a mixture of actual work environment acquisition and point cloud data augmentation, with a total of 658 welded joint workpiece point clouds. Among them, 94 raw point clouds were acquired and screened on-site by a laser vision sensing system, 376 point clouds were acquired by traditional data augmentation based on the raw point clouds, and 188 point clouds were acquired by PointWOLF.

Based on the welding requirements for each welded joint shown in Fig. 1, we labeled the created experimental dataset using the point cloud processing software CloudCompare. During the labeling process, we make the width of the labeled weld area point cloud just enough to accurately cover the welding path of the subsequent intelligent welding robot system, which is an important prerequisite for the subsequent realization of accurate weld area segmentation of the weld zone point cloud and path planning. Meanwhile, for different welded joints, due to their different structural and application requirements in the welding field, different welding devices may be required in the welding process. For example, T welded joints usually require higher welding strength to ensure the stability of the connection, which requires the selection of thicker wire or the use of multipass welding to provide a larger

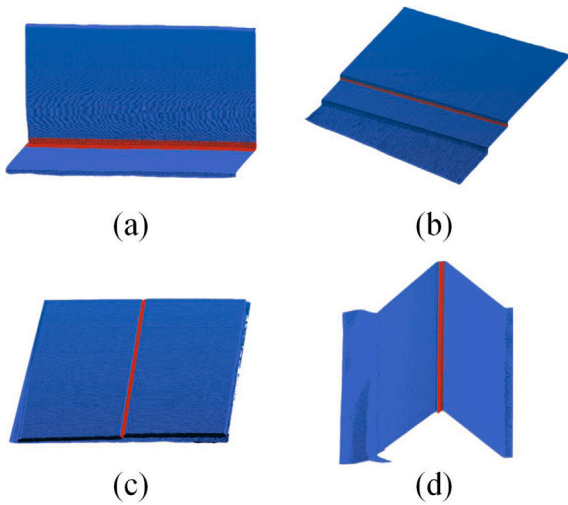


Fig. 3. Schematic of point cloud labeling for common welded joints. (a) T joints. (b) Lap joints. (c) Butt joints. (d) Corner joints.

weld cross-section area, in order to improve the T welded joints of the load-bearing capacity and resistance to deformation. Therefore, in the labeling process, we need to label the point cloud of the weld area according to the application requirements of different welded joints, to facilitate the selection of different welding devices and welding parameters in the subsequent welding process, and the labeling effect of the point cloud of each welded joint is shown in Fig. 3. In a T joint, the two parts intersect at a perpendicular or nearly perpendicular angle, and the point cloud of the weld area is presented as a straight line that extends along the edge of the shorter part and intersects the surface of the longer part. In a lap joint, the two parts partially overlap and the point cloud of the weld area is a line along the overlap area, parallel to the intersection of the two parts. In a butt joint, the two parts are joined end-to-end and the point cloud of the weld area is a straight line along the centerline of the joining surfaces. In a corner joint, the two components intersect at an angle, and its weld area point cloud appears as a curve extending along the corner of the intersection.

Since the point cloud data of the workpiece acquired in the actual working environment is relatively rough, some point cloud data preprocessing operations are needed before and after point cloud data augmentation. Point cloud data are often acquired in different coordinate systems, and thus have different scales and ranges. To reduce the sensitivity of the network model to the point cloud scale and coordinate system, and to improve the generalization ability of the network model, we carry out point cloud data mean normalization on the acquired raw point cloud data before point cloud data augmentation. Influenced by the size of the workpiece and the environment, the amount of point cloud data collected by the laser vision sensing system is relatively large, which is computationally intensive and time-consuming to process directly. Therefore, after completing the labeling of all the point cloud data, we resample the labeled point cloud data using farthest point sampling, which reduces the amount of computation while preserving the key features and structural information in the point cloud dataset.

4. The proposed LWSNet

For the real-time and accuracy requirements of welding tasks in industrial production, we propose a novel point cloud part segmentation network model. In this section, the general framework of LWSNet is first introduced, and then the key constituent modules of LWSNet, including the Feature Extraction Block (FEB), the Feature Generation Block (FGB), and the Feature Propagation Block (FPB), are presented separately.

4.1. General framework of LWSNet

PointNet++ is a deep learning network architecture for processing 3D point cloud data. It is an extended and improved version of PointNet. Compared with the previous PointNet network, the PointNet++ network improves the processing ability of point cloud data by introducing local feature extraction and multi-scale feature learning. This paper selects the part segmentation network model of PointNet++ as the research object, and its network model framework is shown in Fig. 4. The core structure of the PointNet++ part segmentation network model lies in its hierarchical Set Abstraction. The so-called hierarchical means using multiple Set Abstraction modules for feature extraction. The difference is that the number of sampling points and sampling radius of each Set Abstraction module are different. As the level deepens, the number of sampling points and sampling radius will increase accordingly.

The general framework of our proposed LWSNet is shown in Fig. 5. Compared to the PointNet++ part segmentation network model, we increase the number of layers of the network model to better extract features at different layers of the point cloud data. In the PointNet++ part segmentation network model, to enhance the expressive ability of the model, the network model uses MLP to extract the point cloud features in the feature extraction phase, but a large number of convolutional operations do not significantly improve the accuracy, and they also lead to a decrease in the inference speed of the network model, and a large increase in the number of parameters and the amount of computation. Therefore, we lighten the network model, and in the feature extraction module, we use only one depthwise separable convolution for feature extraction. At the same time, we redesigned the Feature Extraction Block (FEB) and the Feature Generation Block (FGB) to obtain better global feature information that incorporates local features.

The point cloud processing flow of LWSNet can be divided into two stages: down-sampling and up-sampling. Specifically, for an input point cloud, in the downsampling stage, the FEB will sample and group the points in the point cloud to obtain the feature information of each neighboring point in the point cloud as well as its relative positional information with respect to the sampled points, and then perform preliminary feature extraction on the fused feature information. The FGB will extract local feature information from the point cloud feature information generated in the FEB to generate global feature information that fuses the local features. In the downsampling stage, four times of global feature information will be generated, and the dimension of the feature information increases with the deepening of the downsampling. After that, we enter the upsampling stage, where the FPB fuses the high-dimensional feature information generated in the downsampling stage inversely layer by layer, and the last FPB is fused with the feature information of the original point cloud data, thus obtaining the fused feature information of each point in the point cloud data. After that, each point in the point cloud data is classified by the network model detection head LWS Head, and the segmented point cloud effect is obtained.

4.2. Feature extraction block

To accurately and completely extract the feature information of each point in the point cloud as well as the local feature information of the point cloud, we designed the FEB module, as shown in Fig. 6. The Sampling&Grouping module is designed with reference to PointNet++, and uses the farthest-point sampling algorithm and the spherical region searching method, respectively. The introduction of the depthwise separable convolution (DSCnv) module greatly reduces the amount of parameters and calculations of the module and makes the network structure more lightweight; the introduction of the normalization-based attention module (NAM) enables the module to focus on the significant features in the channel and extract more accurately local feature information of point cloud.

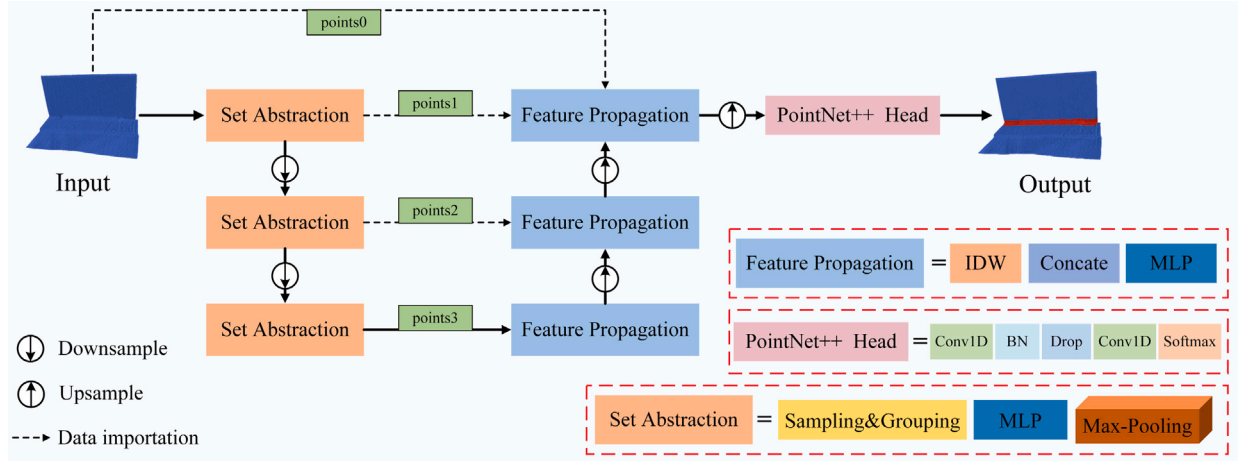


Fig. 4. PointNet++ part segmentation network model framework.

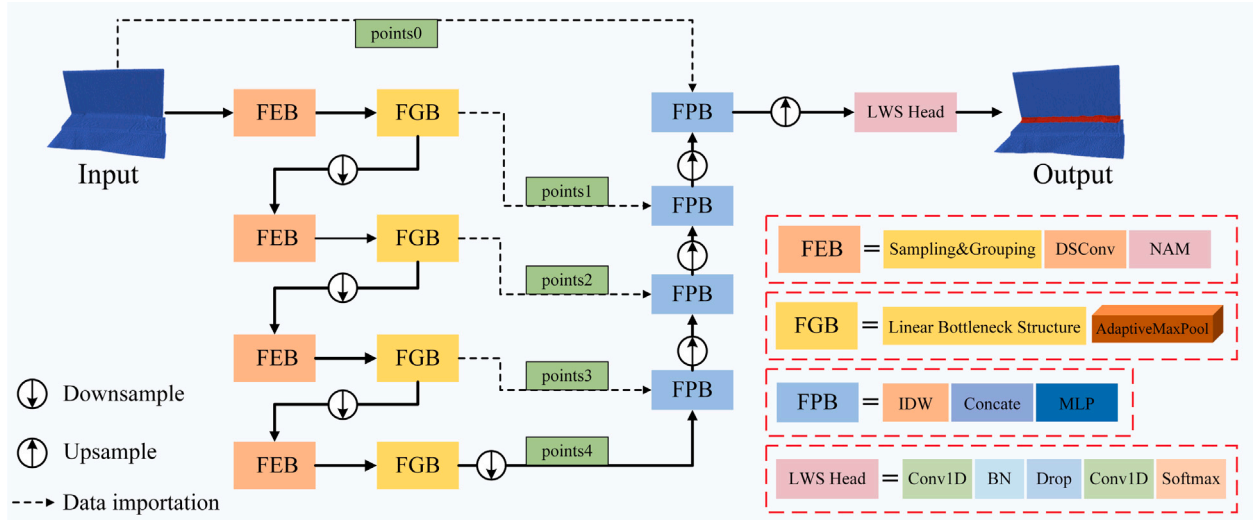


Fig. 5. Overall framework of the proposed model. LWSNet is improved based on the PointNet++ part segmentation model to give the network model a more lightweight structure and better segmentation capability.

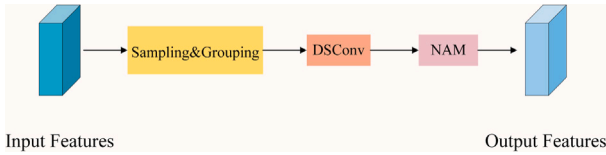


Fig. 6. The proposed point cloud feature extraction block (FEB), FEB consists of Sampling&Grouping module, Depthwise Separable Convolution module, NAM Attention Mechanism and others.

4.2.1. Depthwise separable convolution

Depthwise separable convolution [20] is a special kind of convolution operation that plays an important role in our lightweight network model design. Depthwise separable convolution consists of two steps: depthwise convolution (DW) and pointwise convolution (PW). First, a separate spatial convolution operation, called depthwise convolution, is performed for each channel of the input. This step spatially extracts features from the image, preserving the information independence between channels. Then, the output of the deep convolution is linearly combined and compressed between channels using a 1×1 convolution kernel, called pointwise convolution. This step combines and compresses the features over the channels, reducing the dimensionality of the feature map. Compared to standard convolution,

deep separable convolution separates the spatial convolution from the channel convolution, thus reducing the amount of computation and the number of parameters.

Assume that the size of the input feature map is $m \times h \times w$ where m , h , and w correspond to the number of channels, height, and width of the input feature map, respectively, and the size of the output feature map is $n \times h \times w$ where n is the number of channels of the output feature map. A comparison of standard convolution and depthwise separable convolution is shown in Fig. 7. The expression for the parametric quantitative comparison between the standard convolution and the depthwise separable convolution is:

$$\frac{P_D}{P_S} = \frac{k \times k \times 1 \times m + 1 \times 1 \times m \times n}{k \times k \times m \times n} = \frac{1}{n} + \frac{1}{k^2} \quad (3)$$

The expression for the comparison of the computational effort of the standard convolution and the depthwise separable convolution is:

$$\frac{C_D}{C_S} = \frac{k \times k \times m \times h \times w + 1 \times 1 \times m \times n \times h \times w}{h \times w \times k \times k \times m \times n} = \frac{1}{n} + \frac{1}{k^2} \quad (4)$$

Where P_D denotes the number of parameters of the standard convolution, P_S denotes the number of parameters of the depthwise separable convolution, C_D denotes the computational amount of the standard convolution, and C_S denotes the computational amount of the depthwise separable convolution. From this comparison, it can be seen that

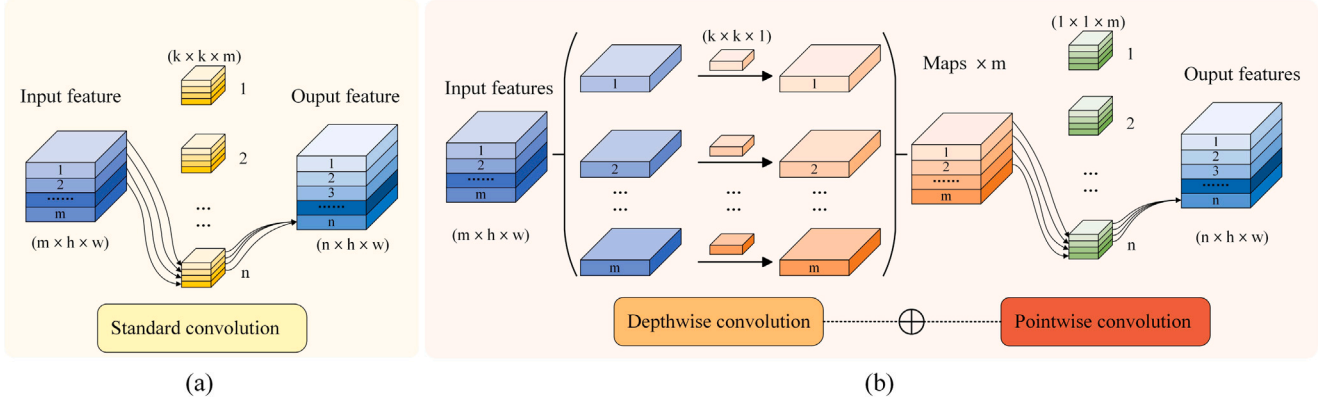


Fig. 7. Comparison of standard convolution and depthwise separable convolution. (a) Standard convolution. (b) Depthwise separable convolution.

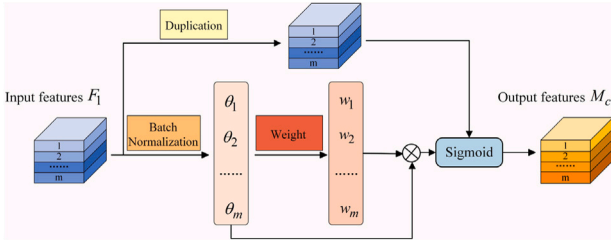


Fig. 8. NAM channel attention submodule.

the number of parameters and the computational amount of the depthwise separable convolution will be significantly reduced compared to the standard convolution.

4.2.2. NAM attentional mechanisms

The attention mechanism can help neural networks suppress less salient features in the channel or the space. Many researches focus on how to obtain salient features by attention operators, and these researches successfully discover the amount of mutual information between different dimensions of the features, but they all lack the consideration of the contribution factor of the weights, while the contribution factor of the weights can further inhibit the information of non-significant features. NAM [21] aims to use the contribution factor of the weights to enhance the effect of the attention by using Batch Normalization's scaling factor to represent the importance of weights, proposing an efficient and lightweight attention mechanism. NAM adopts the module integration approach of CBAM [43] by redesigning the channel and spatial attention submodules.

In this paper, we use the channel attention submodule, which is inserted behind the DSConv module in each layer, to reduce the weight of less significant point cloud features, so that the network model can extract the local features of the point cloud more accurately, the structure of this module is shown in Fig. 8, where F_1 denotes the input features, M_c denotes the output features, and θ_i is the scaling factor of each channel. The weights w_i are calculated as:

$$w_i = \frac{\theta_i}{\sum_{j=0} \theta_j} \quad (5)$$

The output feature M_c is calculated as:

$$M_c = \text{sigmoid}(W_\theta(BN(F_1))) \quad (6)$$

The Channel Attention submodule reflects the size of each channel change as well as its significance through the scaling factor in Batch Normalization, which is calculated as follows:

$$\theta = B_{out} = BN(B_{in}) = \gamma \frac{B_{in} - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta \quad (7)$$

where μ_B and σ_B^2 are the mean and standard deviation of mini batch B , respectively; γ and β are trainable affine transformation parameters representing scale and displacement, respectively.

4.3. Feature generation block

Inspired by [44], we designed the Feature Generation Block FGB, as shown in Fig. 9. The point cloud feature information extracted from the FEB module is sequentially subjected to 1×1 pointwise convolution, 3×3 depthwise convolution, and 1×1 pointwise convolution, in which we use the identity mapping as the activation function during the pointwise convolution operation and sum the obtained feature information with the identity mapping of the original feature information. The generated fused feature information is processed by the adaptive maximum pooling layer to obtain the global feature information that fuses the local features.

4.3.1. Linear bottleneck structure

In [44], the set of activation functions at each layer of a deep neural network is referred to as the “manifold of interest”, and ReLU can only retain complete information about the input manifold if the input manifold is located in a low-dimensional subspace of the input space. Therefore, we use the identity mapping as the activation function instead of the original nonlinear activation transformation in many places in the FGB module to prevent the nonlinearity from destroying too much information. Meanwhile, to improve the ability of gradient propagation between multiplication layers, we make a shortcut connection between two linear bottleneck layers in the FGB module. The design of this linear bottleneck structure can effectively solve the problem of gradient vanishing in the network model and reduce the information loss when the feature information is transmitted between different dimensional spaces.

4.3.2. Adaptive maximum pooling

In the part about global feature extraction, PointNet++ and PointNet have the same practice of aggregating global features through the maximum pooling operation; although PointNet++ achieves step-by-step downsampling through multiple Set Abstraction modules, and obtains different scales and levels of local global features, which makes the final global feature expression ability of the obtained features. Although PointNet++ achieves local global features of different scales and levels through multiple Set Abstraction modules, which makes the global feature expression ability of the final obtained features enhanced, it still causes part of the information loss, and the ability of the network model in local detail processing still needs to be improved.

We use the adaptive maximum pooling layer in the FGB module for global feature generation, which enables the network model to handle input feature maps of different sizes more flexibly and improves

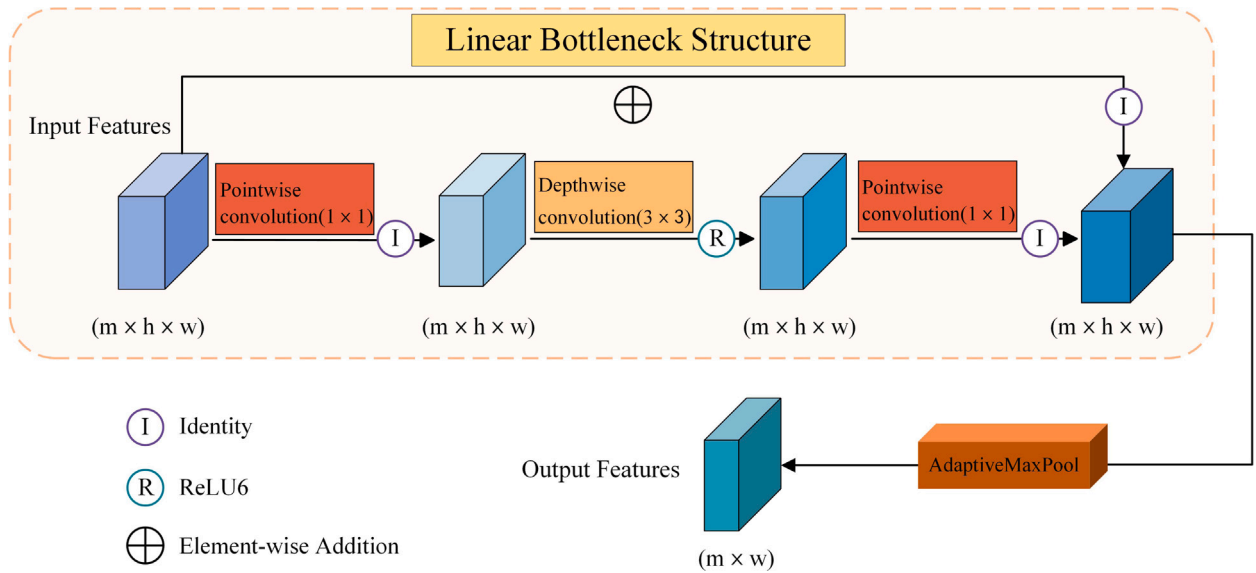


Fig. 9. The proposed feature generation block (FGB), FGB consists of linear bottleneck structure, adaptive maximum pooling layer and other components.

the network's representation and generalization capabilities. Compared with max-pooling, adaptive max-pooling can flexibly adjust the size of the output feature map as needed to adapt to different sizes of input feature maps and network structures. At the same time, adaptive max-pooling does not need to specify a fixed-size pooling window, which is more suitable for processing input feature maps with large size variations and avoids the information loss that may be brought by a fixed-size pooling window. In the FGB of our network model, since the scale of the point cloud feature information captured by each FEB module is different, the use of the adaptive maximum pooling layer can extract global features more flexibly and completely.

4.4. Feature propagation block

To effectively propagate the local features of the point cloud learned in the upsampling phase of our network model to the whole point cloud, we designed the upsampling module FPB. The upsampling module Feature Propagation of PointNet++ does not depend on the specific network structure, and it can be adapted or extended according to the needs of different applications, with good generalization ability, and also has fewer computational parameters as well as higher computational efficiency. Therefore, our FPB module is designed based on Feature Propagation, an upsampling module of PointNet++, and its specific structure is shown in Fig. 10. The IDW module represents the interpolation calculation to generate the interpolated point cloud features through the inverse weight distance method; the MLP module represents the multi-layer perceptron, which extracts the feature information of the fused point cloud features.

5. Experimental results and analysis

5.1. Experimental environment

The experiments listed in this paper were conducted in the Pytorch deep learning framework installed under the Windows operating system, with a software environment of Python 3.8, CUDA 11.6, Pytorch 1.13.1, a processor of 12th Gen Intel(R) Core(TM) i7-12700H@2.30 GHz, and a video card of NVIDIA GeForce RTX 3060.

To achieve point cloud data acquisition for four different welded joint workpieces, we designed a point cloud data-driven laser vision intelligent welding robot system. The robot used in this system is the JAKA Zu 7 robot produced by JAKA Robotics Co., Ltd. The experimental platform of this system is shown in Fig. 11.

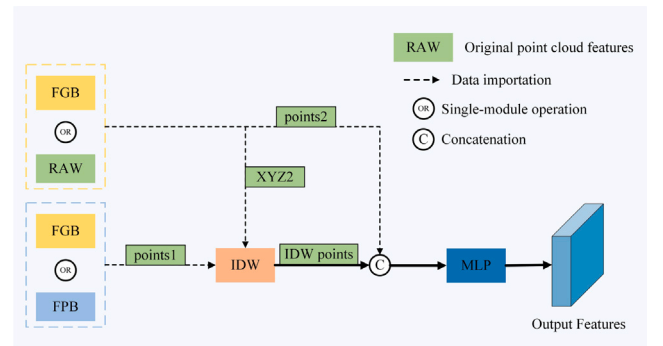


Fig. 10. The proposed feature propagation block (FPB), FPB is designed based on the upsampling module Feature Propagation of PointNet++.

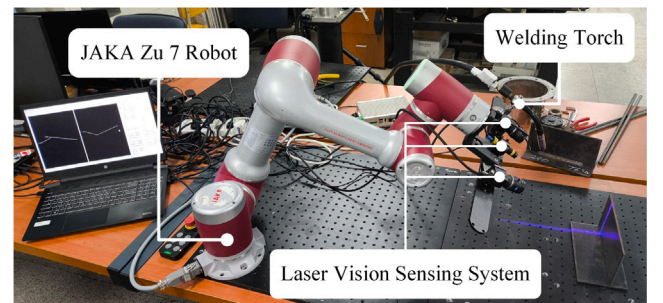


Fig. 11. Laser vision intelligent welding robot experimental platform.

The laser vision sensing system is mounted on the end of the robot's arm in an "eyes on hand" manner and is used to acquire the point cloud of the workpiece. Firstly, a line laser scanning is performed on the welded joint workpiece. To maintain the consistency between the laser streak map acquired by the camera and the robot's motion data, we use multi-threaded control of the camera's stream fetching operation and the robot's position saving operation, and condition variables are used to control the synchronized execution of the threads. When constructing the WeldJoint-PCD dataset, to ensure the quality of the collected point cloud data, for a welded joint with a length of about 200 mm, we set the robot's movement step speed to 5 mm/s,

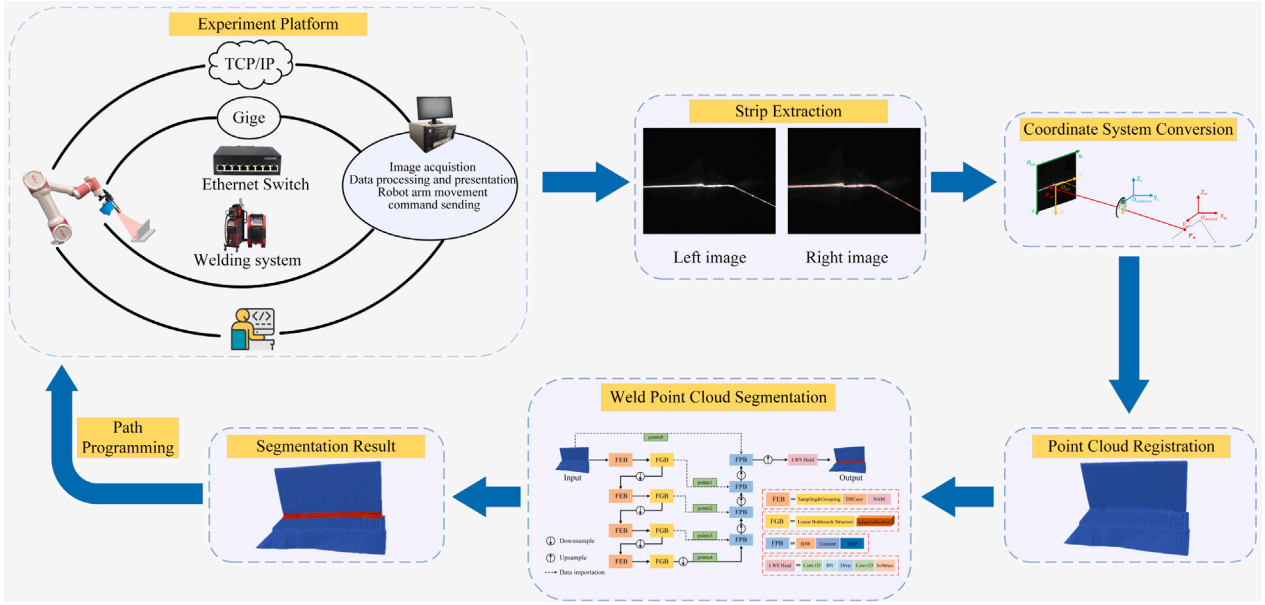


Fig. 12. Block diagram of the structure of the laser vision intelligent welding robot system, showing the detailed workflow of the point cloud data-driven laser vision intelligent welding robot system.

and the number of points in the collected individual point cloud data is about 600 000. The scanned laser stripe image is transmitted to the industrial computer through the switch for processing. Then the center line of the light strip in the laser stripe image is extracted for coordinate transformation and splicing to obtain a complete three-dimensional point cloud. Then, the weld area point cloud segmentation operation is performed through the network model we proposed to obtain the weld area point cloud information. Finally, path planning is performed and the welding path information is sent to the robot to enable it to perform the welding task. The system structure block diagram is shown in Fig. 12.

During the training process, the dataset is divided into training set, validation set and test set in the ratio of 8:1:1. Adam is used as the optimizer of the model, the initial learning rate is 0.001, the weight decay is set to $1e-4$, the Batch size is set to 16, the normal information of the point cloud is used, and the number of network iterations for training is 200, and the learning rate is decayed to 0.5 times of the previous learning rate every 20 training cycles.

In the next experiments in Sections 5.2 and 5.3, we use the number of parameters (Params), the number of floating-point computations per second (FLOPs), the inference time (IT), the categorization intersection and merger ratio (cls.mIoU), and the instance segmentation intersection and merger ratio (ins.mIoU) as evaluation metrics. The number of parameters (Params), floating point computations per second (FLOPs) and inference time (IT) are used to evaluate the lightweighting effect of the model, where we test the computation of FLOPs with a single batch of tensor and the computation of IT using the same point cloud test data. The segmentation effect of the weld area point cloud is evaluated using the classification intersection and merger ratio (cls.mIoU), and the instance segmentation intersection and merger ratio (ins.mIoU). Among them, mIoU is a commonly used performance evaluation metric in the instance segmentation task to measure the similarity between the predicted results and the real labels. It can synthesize the prediction accuracy of each category and provide an overall evaluation result. The calculation formula of mIoU is as follows:

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_{j=0}^k p_{ij} + \sum_{j=0}^k p_{ji} - p_{ii}} \quad (8)$$

where k denotes the number of categories and $k+1$ denotes the number of categories plus the background class, i represents the true value, j represents the predicted value, p_{ii} represents predicting i as i , which is

a true positive (TP), p_{ij} represents predicting i as j , which is a false negative (FN), and p_{ji} represents predicting j as i , which is a false positive (FP).

5.2. Ablation experiments

To analyze the impact of each improvement module on the model performance, we conduct ablation experiments based on the WeldJoint-PCD dataset. We set the proposed network model backbone “backbone” as the basic lightweight model obtained by replacing adaptive max-pooling with max-pooling on the basis of LWSNet without using the NAM attention mechanism and without using linear bottleneck structure. “backbone-NAMA” indicates that the NAM attention mechanism is added to “backbone”. “backbone-LBS” denotes a linear bottleneck structure added to “backbone”. “backbone-AMP” denotes the use of an adaptive maximum pooling layer instead of a normal maximum pooling layer on top of “backbone”. Our proposed network model has the highest cls.mIoU and ins.mIoU compared to other incomplete network models, which means that our complete network model can have better point cloud segmentation compared to other incomplete network models.

The ablation experiment results are shown in Table 2. The experimental data show that the cls.mIoU and ins.mIoU of the original PointNet++ part segmentation network model are relatively low, and Params, FLOPs and IT are high. Compared with the original PointNet++ part segmentation network, the cls.mIoU and ins.mIoU of “backbone” are increased by 2.3% and 3.6% respectively, and the Params, FLOPs and IT of the network model are reduced by 54.3%, 88.6% and 11.5% respectively. It can be seen that the obtained lightweight model has greatly improved the segmentation effect and lightweight degree. Based on “backbone”, each module component test “backbone-NAMA”, “backbone-LBS” and “backbone-AMP” are carried out. The experimental results show that each module component has improved the segmentation ability of the network model to varying degrees. At the same time, the Params and FLOPs of the network model only increase slightly and IT does not fluctuate significantly. Compared with the original PointNet++ part segmentation network model, “Ours” improves cls.mIoU and ins.mIoU by 4.2% and 5.7% respectively, and reduces Params, FLOPs and IT by 33.4%, 76.0% and 10.0% respectively, proving that our proposed network model has superior comprehensive performance.

Table 2

Ablation experiments based on WeldJoint-PCD dataset.

Methods	cls.mIoU (%)	ins.mIoU (%)	Params (M)	FLOPs (G)	IT (ms)
PointNet++	88.9	86.4	1.734	4.91	511.0
backbone	91.2	90.0	0.792	0.56	452.0
backbone-NAMA	91.6	90.2	0.794	0.58	467.1
backbone +LBS	92.2	91.2	1.153	1.17	455.0
backbone +AMP	92.6	91.8	0.792	0.56	441.0
Ours	93.1	92.1	1.155	1.18	460.0

Table 3

Comparison experiments based on the WeldJoint-PCD dataset.

Methods	cls.mIoU (%)	ins.mIoU (%)	Params (M)	FLOPs (G)	IT (ms)
PointNet	74.1	66.8	8.334	5.76	188.5
PointNet++	88.9	86.4	1.734	4.91	511.0
PointMLP	90.2	88.5	15.295	6.05	512.0
DGCNN	91.5	92.0	1.454	5.03	467.0
Point Transformer	75.1	70.5	19.394	146.97	733.0
PAConv	90.0	89.2	1.058	1.95	472.0
Ours	93.1	92.1	1.155	1.18	460.0

5.3. Comparison experiments

In this section, to prove that our proposed LWSNet network model has good comprehensive performance in weld point cloud segmentation, we conduct a comparison experiment between LWSNet and the current popular point cloud part segmentation network model based on the WeldJoint-PCD dataset.

The results of the comparison experiments based on the WeldJoint-PCD dataset are shown in Table 3, and the experimental data show that the LWSNet network model proposed in this paper has the best weld point cloud segmentation, reaching 93.1% and 92.1% in cls.mIoU and ins.mIoU, respectively. Meanwhile, the Params, FLOPs, and IT of LWSNet reach the combined best compared to other network models in the comparison experiments, proving that our network model has the best lightweight level. Compared with multilayer perceptron (MLP)-based point cloud part segmentation network models such as PointNet, PointNet++, and PointMLP [45], our network model shows better performance in terms of segmentation effectiveness and lightweight degree. Compared with graph convolution based part segmentation network models such as DGCNN [46], our network model has better performance in terms of segmentation effect and degree of lightweighting. Compared with self-attention mechanism based part segmentation network models such as Point Transformer [47], our network model has superior performance in terms of segmentation effect and lightweight degree. Compared to point convolution based part segmentation network models such as PAConv [48], the PAConv network model also has a high degree of lightweighting, while our network model achieves better segmentation results. Figs. 13 and 14 show the changes in the cls.mIoU and ins.mIoU of each network model during the training period, respectively.

5.4. Detection effect and result analysis

To verify the detection effect of our proposed network model in practical applications, we select some original point clouds collected and screened by the laser vision sensing system on-site and the point clouds enhanced by point cloud data to add noise to simulate the strong noise environment in the industrial field to form a test set. The trained network model in Section 5.3 is used to test the point clouds in the test set for weld seam area point cloud segmentation, and a comparison of the detection effect of the original point clouds acquired and screened by the laser vision sensing system in the field is shown in Fig. 15, and a comparison of the detection effect of the point clouds augmented by point cloud data with added noise to simulate the strong noise

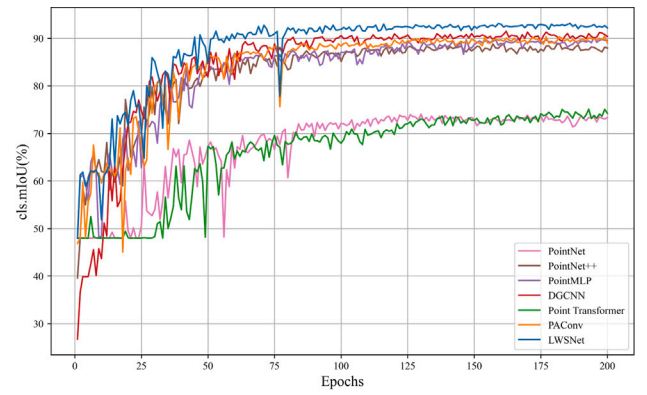


Fig. 13. Comparison of changes in cls.mIoU during training of each network model based on WeldJoint-PCD dataset.

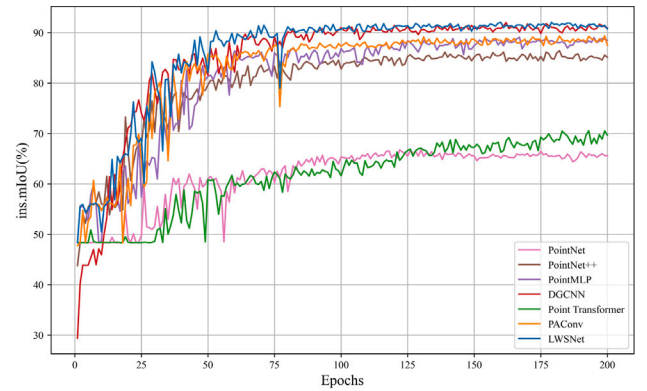


Fig. 14. Comparison of ins.mIoU changes during training of each network model based on WeldJoint-PCD dataset.

environment in the industrial field is shown in Fig. 16. To present the advantages of our proposed network model more clearly, some details of the detection results are enlarged.

The experimental results show that all the network models used for comparison are much better in detecting the part of the original point cloud for T and corner joint workpieces compared to lap and butt joint workpieces, but there are still a small number of false detections. PointNet is almost unable to detect the weld area point cloud of lap and butt joint workpieces. PointMLP, DGCNN, and Point Transformer all show varying degrees of false detections on T joint workpieces, and PointNet++ and PAConv can show relatively good detection on T and corner joint workpieces. Meanwhile, all the network models used for comparison show different degrees of false detection and missed detection for lap and butt joint workpieces, especially in the detection of butt joint workpieces. In contrast, our proposed network model shows good detection results for all types of welded joint workpieces.

In simulating the partial detection effect of the point cloud obtained at the industrial site, it can be seen that the noise affects all network models to varying degrees. In all the network models used for comparison, there is false detection of noise in addition to varying degrees of false detection and missed detection on the point cloud of the workpiece. In contrast, our proposed network model has no false detection of noise on all types of welded joints, and the noise has less impact on the detection of the workpiece point cloud itself. Overall, our proposed network model shows good detection results as well as anti-interference ability.

Generally, the network model we proposed can accurately identify the weld area point clouds in different welding joint workpieces on

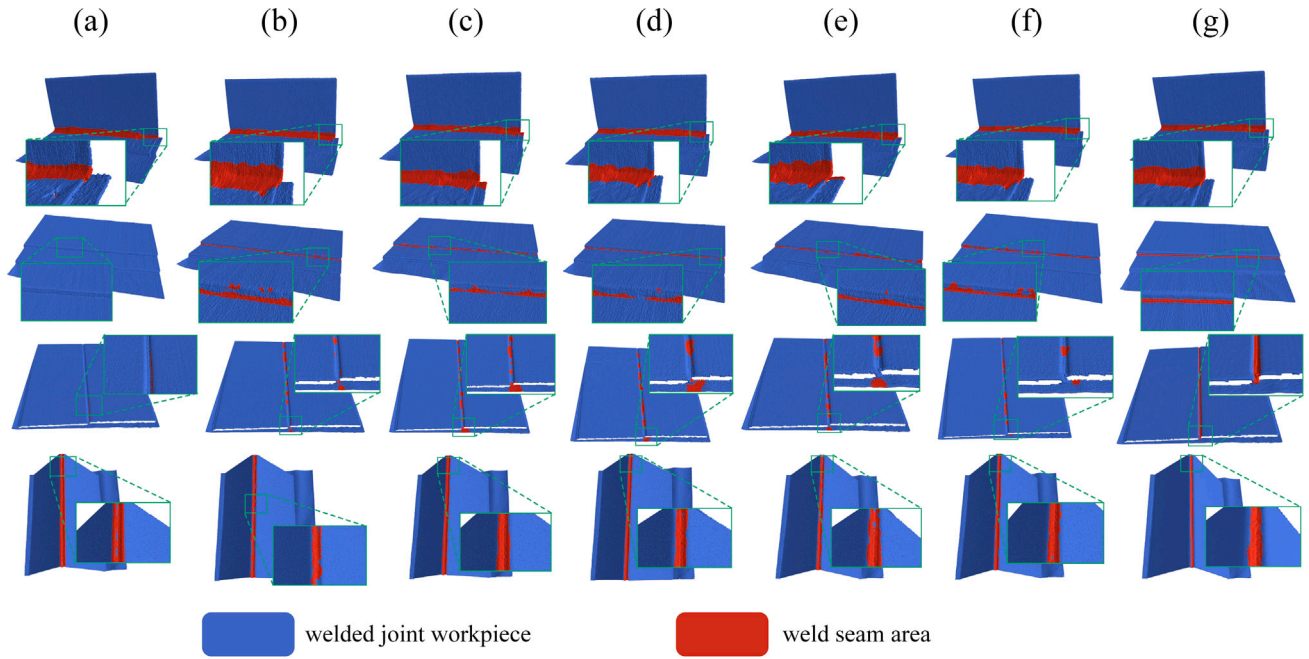


Fig. 15. Comparison of the visualized detection results of each network model on the original point cloud test set, with some of the detection details zoomed in. (a) PointNet. (b) PointNet++. (c) PointMLP. (d) DGCNN. (e) Point Transformer. (f) PACov. (g) Ours.

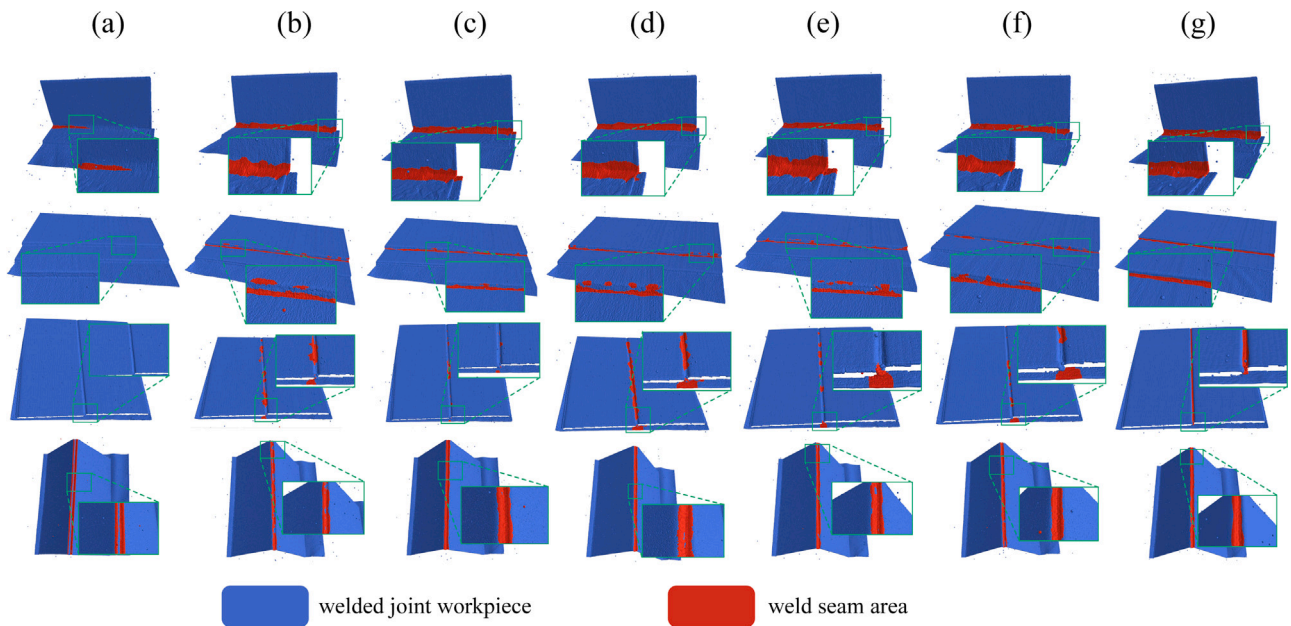


Fig. 16. Comparison of the visualization of the detection effects of each network model on a simulated industrial field point cloud test set, with some of the detection details zoomed in. (a) PointNet. (b) PointNet++. (c) PointMLP. (d) DGCNN. (e) Point Transformer. (f) PACov. (g) Ours.

both the original point cloud test set and the simulated industrial site point cloud test set, and can provide clear weld boundaries and feature points, which is suitable for subsequent path planning and guiding the robot to perform welding tasks.

5.5. Model generalization validation

5.5.1. Verification experiment based on semi-welded joint

Considering that not all welding tasks in industrial sites are directly aimed at unwelded welded joint workpieces, such as T welded joints,

it is difficult to keep T welded joints in the unwelded state to be welded, and their vertical parts are difficult to remain stable. Often in industrial sites, partial welding is performed manually before the welding robot completes the remaining welding tasks. Therefore, the weld segmentation model should have the ability to segment the point cloud of the weld area of a semi-welded workpiece for subsequent welding work by a welding robot.

In this paper, some semi-welded T welded joint workpiece point clouds are acquired, and the semi-welded T welded joint workpiece is shown in Fig. 17. The point cloud of the original semi-welded T

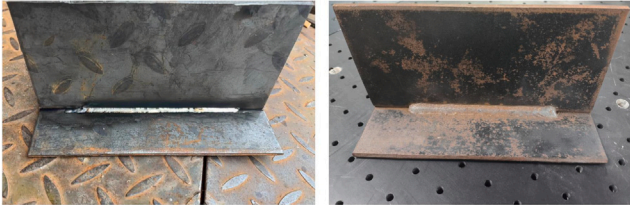


Fig. 17. Semi-welded T joint workpiece.

joint workpiece is added with noise by point cloud data augmentation method to simulate the point cloud obtained under the strong noise environment in the industrial field. Using the trained network models in Section 5.3, the two point clouds are tested for weld segmentation, and the results of the weld segmentation test are shown in Fig. 18. To present the advantages of our proposed network model more clearly, some of the details of the detection results are enlarged.

The results of the weld segmentation test of the point cloud of a semi-welded T joint show that in the original detection of the point cloud of a semi-welded T joint, all the network models used for the comparison have different degrees of misdetection and omission of the point cloud of the weld seam area. Meanwhile, the addition of noise to simulate the strong noise environment in the industrial site further aggravates the degree of misdetection and omission of the point cloud in the weld area of the network models used for comparison, and the misdetection of the noise occurs. However, our network model shows good detection results both in the original semi-welded T joint workpiece point cloud detection and in the point cloud detection obtained under the simulated industrial site noise environment, which indicates that our network model has good anti-interference ability and generalization ability on the unknown point cloud data.

5.5.2. Validation experiments based on ShapeNetPart

Our proposed network model is obtained by improving based on the PointNet++ part segmentation network model. In Section 5.3, we conducted a comparison experiment with the current more popular point cloud part segmentation network based on the WeldJoint-PCD dataset and verified that our network model exhibits better comprehensive performance compared to the same type of network model. To further validate the generalization of the LWSNet network model, we choose a public dataset for comparison experiments. The dataset used is the point cloud part segmentation dataset ShapeNetPart, which contains 16 object categories and 50 part categories, and contains a total of 16,881 point cloud samples, with 14,007 samples in the training set and 2,874 samples in the test set. We still use the six networks in Section 5.3 for the comparison experiments, and since we have already conducted the tests of each network model Params, FLOPs, and IT in the comparison experiments in Section 5.3, we only use the evaluation metrics of cls.mIoU and ins.mIoU here.

The results of the comparison experiments based on the ShapeNetPart dataset are shown in Table 4, and the experimental data show that our model still exhibits better part segmentation performance on the ShapeNetPart dataset compared to network models such as PointNet, PointNet++ and DGCNN. Compared to the part segmentation network models such as PointMLP, Point Transformer and PAConv, our network model still has some gaps in part segmentation performance.

It is worth noting that our network model is proposed for the weld segmentation task, and the experimental dataset we are oriented to, WeldJoint-PCD, has fewer but more focused and in-depth data types compared to the ShapeNetPart dataset. In the weld segmentation task, to ensure the accuracy of weld segmentation, the network model needs to have a more in-depth understanding of the local and global structure of the point cloud, so our network model has more feature extraction

Table 4

Comparison experiments based on the ShapeNetPart dataset.

Methods	cls.mIoU (%)	ins.mIoU (%)
PointNet	80.4	83.7
PointNet++	81.9	85.1
PointMLP	84.6	86.1
DGCNN	82.3	85.2
Point Transformer	83.7	86.6
PAConv	84.6	86.1
Ours	82.9	85.5

layers than the original network structure. At the same time, to meet the real-time requirements of the weld segmentation task, we do not use MLP or some complex convolutional processing to extract the deep features of the point cloud data as other network structures do. These limitations make our network model may not be suitable to handle the ShapeNetPart-based part segmentation task, but the segmentation performance of our network model still outperforms the original network model, which again shows that our network has a good generalization ability on unknown point cloud data and stability in the face of the changes generated in the point cloud data.

6. Conclusion

In this paper, we propose a novel point cloud part segmentation network model, LWSNet, to realize quickly segmenting the point clouds of weld areas in different welded joints for subsequent path planning and guiding the robot to perform the welding work. Our network model is improved based on the PointNet++ part segmentation network model, which mainly consists of the Feature Extraction Block (FEB), the Feature Generation Block (FGB), and the Feature Propagation Block (FPB). We introduce depthwise separable convolution and NAM attention mechanism to design the FEB module to improve the lightweight degree and feature extraction capability of the network model. We design a linear bottleneck structure in the FGB module and introduce adaptive maximum pooling to reduce the information loss during the transmission of point cloud feature information and improve the feature generation capability of the network model. We designed a laser vision sensor model to acquire the point cloud data of the welded joint workpiece, and constructed WeldJoint-PCD, a point cloud dataset of the welded joint workpiece, through multiple data augmentation methods for the training, detection, and evaluation of the network model. The ablation and comparison experiments on the WeldJoint-PCD dataset show the effectiveness of each improved modular component of our network model as well as its better overall performance compared to other network models, and the experimental results in the model generalization validation show that our network model has good generalization ability and stability. Compared with the original network model, on the WeldJoint-PCD dataset and the ShapeNetPart dataset, LWSNet, while maintaining a high degree of lightweighting, improves the cls. mIoU by 4.2% and 1.0%, and the ins. mIoU by 5.7% and 0.4%, which verifies its excellent point cloud segmentation performance.

The LWSNet network model is proposed for the weld area point cloud segmentation task in automated welding, and it is known from the validation experiments based on ShapeNetPart that LWSNet still has some limitations on the point cloud segmentation task for other target types. Meanwhile, LWSNet is designed with high flexibility, and modules can be flexibly added and deleted to adapt to different application requirements. Therefore, future work can focus on further optimizing the network structure, improving the real-time and adaptability of the network model, and exploring more industrial application scenarios.

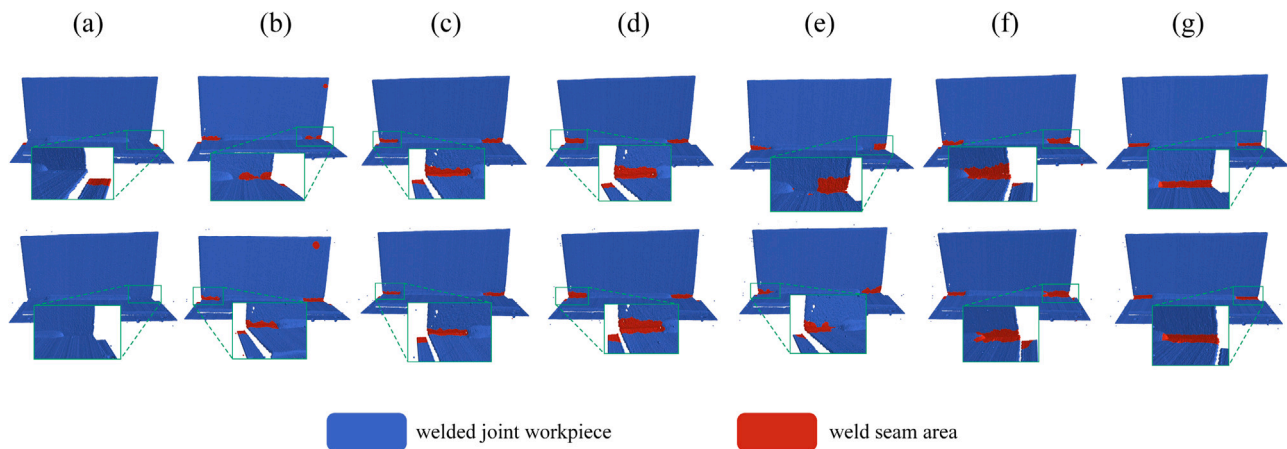


Fig. 18. Comparison of the visualized detection effect of each network model on the point cloud test set of semi-welded T joint workpieces with some of the detection details zoomed in, where the first row is the detection effect of the original semi-welded T joint workpiece point cloud, and the second row is the detection effect of the point cloud of semi-welded T joint workpieces obtained from simulating the strong noise environment of the industrial site. (a) PointNet. (b) PointNet++. (c) PointMLP. (d) DGCNN. (e) Point Transformer. (f) PACConv. (g) Ours.

CRedit authorship contribution statement

Limei Song: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Hongli Wang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Yuezhe Zhang:** Writing – review & editing, Visualization, Validation, Investigation. **Zhi Qiao:** Writing – review & editing, Validation, Supervision, Investigation, Data curation. **Fengyi Han:** Writing – review & editing, Validation, Supervision, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] F. Xu, Y. Xu, H. Zhang, S. Chen, Application of sensing technology in intelligent robotic arc welding: A review, *J. Manuf. Process.* 79 (2022) 854–880.
- [2] T. Lei, Y. Rong, H. Wang, Y. Huang, M. Li, A review of vision-aided robotic welding, *Comput. Ind.* 123 (2020) 103326.
- [3] J. Li, P. Zhou, G. Wei, J. Huang, H. Shi, The microstructure and mechanical properties of titanium/copper welded joint by FSW, *Mater. Sci. Technol.* 38 (17) (2022) 1532–1542.
- [4] Y. Peng, L. Song, L. Zhao, C. Ma, H. Zhao, Z. Tian, Research status of weldability of advanced steel, *Acta Metall. Sin.* 56 (4) (2020) 601–618.
- [5] Y. Xu, Z. Wang, Visual sensing technologies in robotic welding: Recent research developments and future interests, *Sensors Actuators A* 320 (2021) 112551.
- [6] A. Rout, B. Deepak, B. Biswal, Advances in weld seam tracking techniques for robotic welding: A review, *Robot. Comput.-Integr. Manuf.* 56 (2019) 12–37.
- [7] Y. Zou, G. Zeng, Light-weight segmentation network based on SOLOv2 for weld seam feature extraction, *Measurement* 208 (2023) 112492.
- [8] N. Wang, K. Zhong, X. Shi, X. Zhang, A robust weld seam recognition method under heavy noise based on structured-light vision, *Robot. Comput.-Integr. Manuf.* 61 (2020) 101821.
- [9] R. Xiao, Y. Xu, Z. Hou, C. Chen, S. Chen, An adaptive feature extraction algorithm for multiple typical seam tracking based on vision sensor in robotic arc welding, *Sensors Actuators A* 297 (2019) 111533.
- [10] Y. Zou, R. Zhan, Research on 3D curved weld seam trajectory position and orientation detection method, *Opt. Lasers Eng.* 162 (2023) 107435.
- [11] L. Yang, Y. Liu, J. Peng, Z. Liang, A novel system for off-line 3D seam extraction and path planning based on point cloud segmentation for arc welding robot, *Robot. Comput.-Integr. Manuf.* 64 (2020) 101929.
- [12] W. Wang, W. Zhang, X. Liu, X. Zhang, W. Huang, Z. Deng, Point cloud 3D weldment reconstruction and welding feature extraction for robotic multi-bead arc weld cladding path planning, *Int. J. Precis. Eng. Manuf.* (2024) 1–15.
- [13] P. Zhou, R. Peng, M. Xu, V. Wu, D. Navarro-Alarcon, Path planning with automatic seam extraction over point cloud models for robotic arc welding, *IEEE Robot. Autom. Lett.* 6 (3) (2021) 5002–5009.
- [14] Y. Geng, Y. Zhang, X. Tian, X. Shi, X. Wang, Y. Cui, A method of welding path planning of steel mesh based on point cloud for welding robot, *Int. J. Adv. Manuf. Technol.* 116 (2021) 2943–2957.
- [15] Y. Geng, Y. Zhang, X. Tian, X. Shi, X. Wang, Y. Cui, A novel welding path planning method based on point cloud for robotic welding of impeller blades, *Int. J. Adv. Manuf. Technol.* 119 (11) (2022) 8025–8038.
- [16] X. Wang, T. Chen, Y. Wang, D. Zheng, X. Chen, Z. Zhao, The 3D narrow butt weld seam detection system based on the binocular consistency correction, *J. Intell. Manuf.* 34 (5) (2023) 2321–2332.
- [17] O. Jing, Z. Lai, W. Qinghong, L. Xin, L. Yingjie, Weld-seam identification and model reconstruction of remanufacturing blade based on three-dimensional vision, *Adv. Eng. Inform.* 49 (2021) 101300.
- [18] L. Yang, E. Li, T. Long, J. Fan, Z. Liang, A novel 3-D path extraction method for arc welding robot based on stereo structured light sensor, *IEEE Sens. J.* 19 (2) (2018) 763–773.
- [19] C.R. Qi, L. Yi, H. Su, L.J. Guibas, Pointnet++: Deep hierarchical feature learning on point sets in a metric space, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [20] A.G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, H. Adam, Mobilenets: Efficient convolutional neural networks for mobile vision applications, 2017, arXiv preprint [arXiv:1704.04861](https://arxiv.org/abs/1704.04861).
- [21] Y. Liu, Z. Shao, Y. Teng, N. Hoffmann, NAM: Normalization-based attention module, 2021, arXiv preprint [arXiv:2111.12419](https://arxiv.org/abs/2111.12419).
- [22] S. Kim, S. Lee, D. Hwang, J. Lee, S.J. Hwang, H.J. Kim, Point cloud augmentation with weighted local transformations, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 548–557.
- [23] L. Yi, V.G. Kim, D. Ceylan, I.-C. Shen, M. Yan, H. Su, C. Lu, Q. Huang, A. Sheffer, L. Guibas, A scalable active framework for region annotation in 3D shape collections, *SIGGRAPH Asia* (2016).
- [24] Y. He, Y. Xu, Y. Chen, H. Chen, S. Chen, Weld seam profile detection and feature point extraction for multi-pass route planning based on visual attention model, *Robot. Comput.-Integr. Manuf.* 37 (2016) 251–261.
- [25] J. Cheng, H. Jin, X. Qian, Real-time weld seam feature extraction in construction sites, *Autom. Constr.* 160 (2024) 105330.
- [26] X. Chen, Z. He, Q. Ma, Y. Ren, T. Cui, A lightweight efficient semantic segmentation with encoder-decoder for arc interference in robotic arc welding, *Meas. Sci. Technol.* 35 (2) (2023) 025124.
- [27] Y. He, G. Ma, S. Chen, Autonomous decision-making of welding position during multipass GMAW with T-joints: A Bayesian network approach, *IEEE Trans. Ind. Electron.* 69 (4) (2021) 3909–3917.

- [28] N. Banafian, R. Fesharakifard, M.B. Menhaj, Precise seam tracking in robotic welding by an improved image processing approach, *Int. J. Adv. Manuf. Technol.* 114 (1) (2021) 251–270.
- [29] N.F. Johan, H.N. Mohd Shah, M. Sulaiman, O.A.A.M. Naji, M.A. Arshad, Weld seam feature point extraction using laser and vision sensor, *Int. J. Adv. Manuf. Technol.* 127 (11–12) (2023) 5155–5170.
- [30] Y. He, Z. Yu, Y. Deng, J. Deng, R. Cai, Z. Wang, W. Tu, W. Zhong, AHP-based welding position decision and optimization for angular distortion and weld collapse control in T-joint multipass GMAW, *J. Manuf. Process.* 121 (2024) 246–259.
- [31] N. Wang, J. Yang, X. Zhang, T. Gong, K. Zhong, Robust weld seam tracking method based on detection and tracking of laser stripe, *Int. J. Adv. Manuf. Technol.* (2024) 1–13.
- [32] Y. He, R. Cai, F. Dai, Z. Yu, Y. Deng, J. Deng, Z. Wang, G. Ma, W. Zhong, A unified framework based on semantic segmentation for extraction of weld seam profiles with typical joints, *J. Manuf. Process.* 131 (2024) 2275–2287.
- [33] Z. Yuankai, J. Yong, T. Xincheng, X. Xiaolong, G. Yusen, L. Min, A point cloud-based welding trajectory planning method for plane welds, *Int. J. Adv. Manuf. Technol.* 125 (3) (2023) 1645–1659.
- [34] S.M. Ahmed, Y.Z. Tan, C.M. Chew, A. Al Mamun, F.S. Wong, Edge and corner detection for unorganized 3d point clouds with application to robotic welding, in: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2018, pp. 7350–7355.
- [35] J. Kim, J. Lee, M. Chung, Y.-G. Shin, Multiple weld seam extraction from RGB-depth images for automatic robotic welding via point cloud registration, *Multimedia Tools Appl.* 80 (2021) 9703–9719.
- [36] H. Feng, X. Ren, L. Li, X. Zhang, H. Chen, Z. Chai, X. Chen, A novel feature-guided trajectory generation method based on point cloud for robotic grinding of freeform welds, *Int. J. Adv. Manuf. Technol.* 115 (5) (2021) 1763–1781.
- [37] C.R. Qi, H. Su, K. Mo, L.J. Guibas, Pointnet: Deep learning on point sets for 3d classification and segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 652–660.
- [38] V. Vanian, G. Zamanakos, I. Pratikakis, Improving performance of deep learning models for 3D point cloud semantic segmentation via attention mechanisms, *Comput. Graph.* (2022).
- [39] Y. Li, R. Bu, M. Sun, W. Wu, X. Di, B. Chen, Pointcnn: Convolution on x-transformed points, *Adv. Neural Inf. Process. Syst.* 31 (2018).
- [40] C. Szegedy, S. Ioffe, V. Vanhoucke, A. Alemi, Inception-v4, inception-resnet and the impact of residual connections on learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 31, (no. 1) 2017.
- [41] G. Brauwers, F. Frasincar, A general survey on attention mechanisms in deep learning, *IEEE Trans. Knowl. Data Eng.* 35 (4) (2021) 3279–3298.
- [42] S.C. Wong, A. Gatt, V. Stamatescu, M.D. McDonnell, Understanding data augmentation for classification: when to warp? in: 2016 International Conference on Digital Image Computing: Techniques and Applications, DICTA, IEEE, 2016, pp. 1–6.
- [43] S. Woo, J. Park, J. Lee, I.S. Kweon, Chm: convolutional block attention module, in: *Proceedings of the European conference on computer vision (ECCV)*: 3–19, 2018.
- [44] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: Inverted residuals and linear bottlenecks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 4510–4520.
- [45] X. Ma, C. Qin, H. You, H. Ran, Y. Fu, Rethinking network design and local geometry in point cloud: A simple residual MLP framework, 2022, arXiv preprint arXiv:2202.07123.
- [46] Y. Wang, Y. Sun, Z. Liu, S.E. Sarma, M.M. Bronstein, J.M. Solomon, Dynamic graph cnn for learning on point clouds, *ACM Trans. Graph (tog)* 38 (5) (2019) 1–12.
- [47] H. Zhao, L. Jiang, J. Jia, P.H. Torr, V. Koltun, Point transformer, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 16259–16268.
- [48] M. Xu, R. Ding, H. Zhao, X. Qi, Paconv: Position adaptive convolution with dynamic kernel assembling on point clouds, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 3173–3182.